

برای آنالیز از زبان پایتون استفاده شده. از آنجایی که تعداد افراد سالم (GROUP1) کم بود از تکنیک GAN (به فارسی شبکه مولد متخاصم) که یکی از تکنیک های تولید داده بسیار شبیه به داده های واقعی است استفاده شده. (لینک مقاله GAN: [Generative Adversarial Networks](#) [1406.2661])

برای ساختن شبکه GAN از فریمورک PyTorch (یکی از فریمورک های بسیار مطرح در زمینه هوش مصنوعی، منتشر شده توسط متا استفاده شده لینک مقاله : [PyTorch: An Imperative Style, High-Performance Deep Learning Library](#) [1912.01703])

برای رسم نمودارها از کتابخانه Matplotlib استفاده شده است.

برای خواندن دیتاست ها و تولید دیتاست های csv در اکسل از کتابخانه pandas استفاده شده.

برای محاسبه میانگین، انحراف معیار و دیگر محاسبات عددی از کتابخانه numpy استفاده شده.

از الگوریتم K-Means Clustering برای دسته بندی اتوماتیک (بدون نظارت به انگلیسی Unsupervised) برای دسته بندی گروه ۱ و ۲ استفاده شده است تا با اندازه گیری همزمان پارامترهای z ، voxel ، BSC و cop ببینیم آیا تفاوت معناداری بین افراد سالم و مریض در MRI آنها وجود دارد یا خیر لینک مقاله: [\[2304.09989\] CKmeans and FCKmeans : Two deterministic initialization procedures for Kmeans algorithm using a modified crowding distance](#)

برای پیاده سازی الگوریتم KMeans Clustering از کتابخانه sklearn استفاده شده است.

لینک کدهای تولید دیتا با الگوریتم GAN به زبان پایتون:

<https://github.com/ArminKardan/jal/blob/master/gan.py>

لینک کدهای KMeans Clustering:

<https://github.com/ArminKardan/jal/blob/master/kmeans.py>

برای ارزیابی نرمالیتی دیتا از روش Shapiro-Wilk استفاده شده اما به دلیل کمبود دیتاها، دیتا های پرت میتواند نتیجه این الگوریتم را تغییر دهد به همین منظور برای اثبات نرمالیتی دیتا از روش Q-Q استفاده شده است و برای پیاده سازی Q-Q و Shapiro از کتابخانه scipy استفاده شده است.

لینک کدهای Q-Q برای نمایش نرمالیتی:

<https://github.com/ArminKardan/jal/blob/master/QQ.py>

برای بررسی و رسم نمودارهای t-test از کتابخانه scipy استفاده شده است.

لینک کدهای t-test:

<https://github.com/ArminKardan/jal/blob/master/diagram.py>

لینک های دیتاست ها:

گروه ۱ استم چپ (آگمنت شده):

<https://github.com/ArminKardan/jal/blob/master/G1LT-A.csv>

گروه ۱ استم چپ (اورجینال):

<https://github.com/ArminKardan/jal/blob/master/G1LT.csv>

گروه ۲ استم چپ:

<https://github.com/ArminKardan/jal/blob/master/G2LT.csv>

گروه ۱ استم راست (آگمنت شده):

<https://github.com/ArminKardan/jal/blob/master/G1RT-A.csv>

گروه ۱ استم راست (اورجینال):

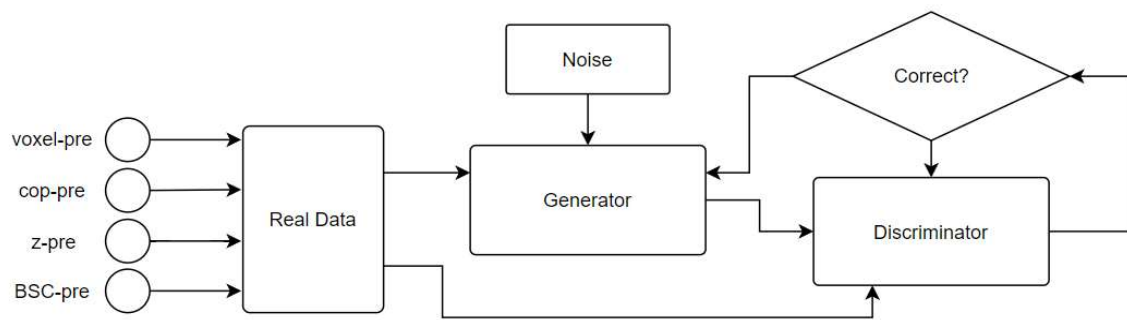
<https://github.com/ArminKardan/jal/blob/master/G1RT.csv>

گروه ۲ استم راست:

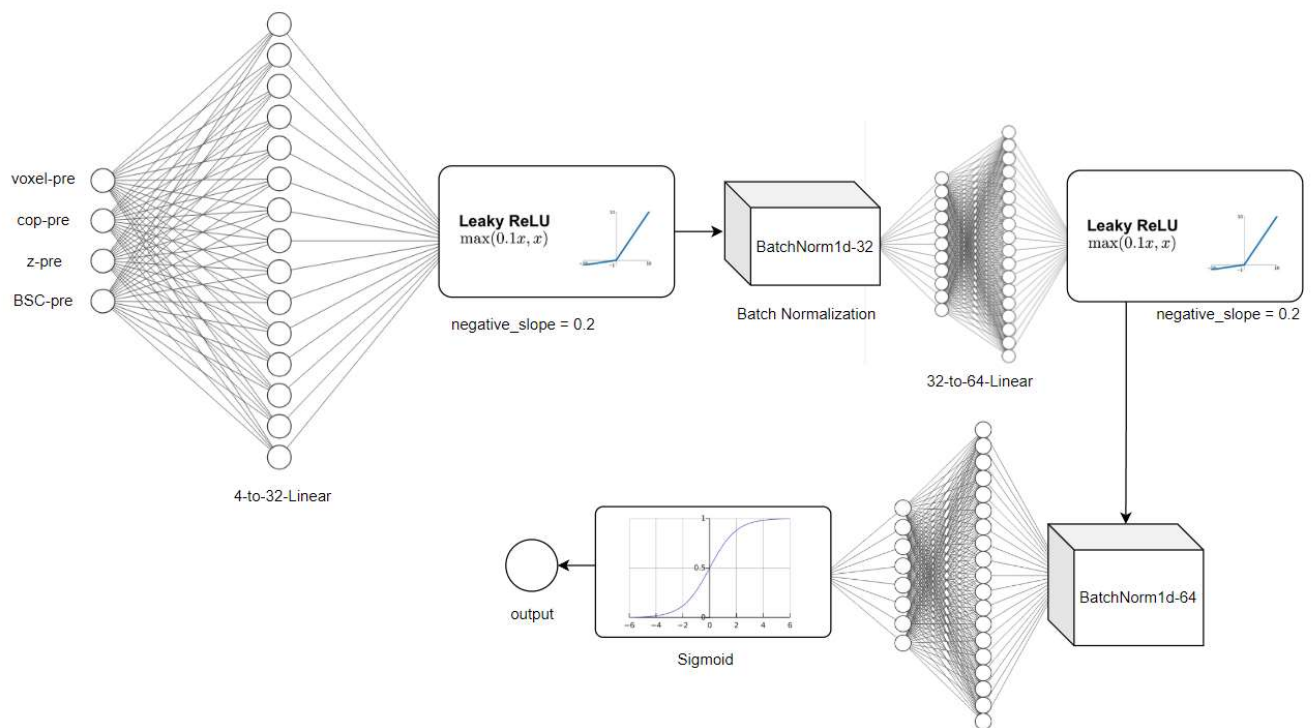
<https://github.com/ArminKardan/jal/blob/master/G2RT.csv>

شاخص ARI:

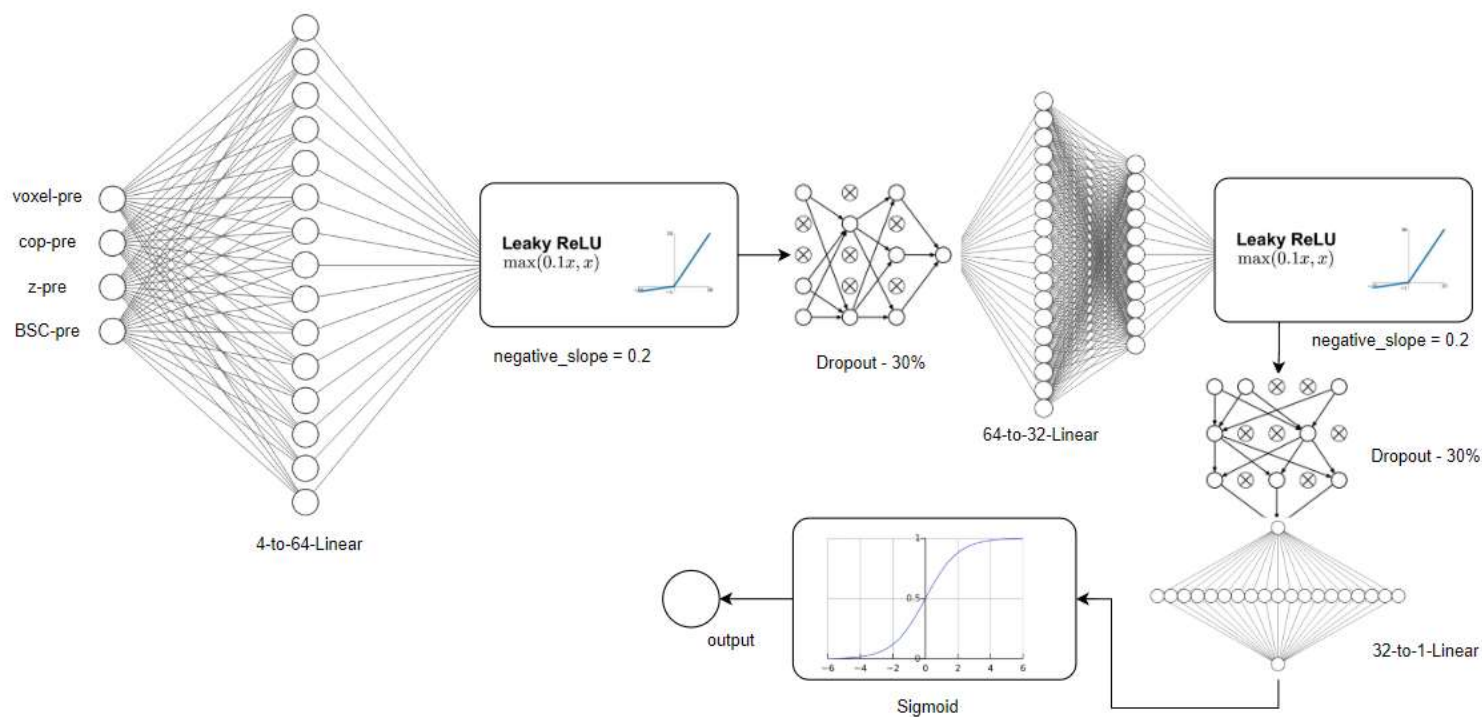
[Understanding the Adjusted Rand Index and Other Partition Comparison Indices Based on Counting Object Pairs | Journal of Classification](#)



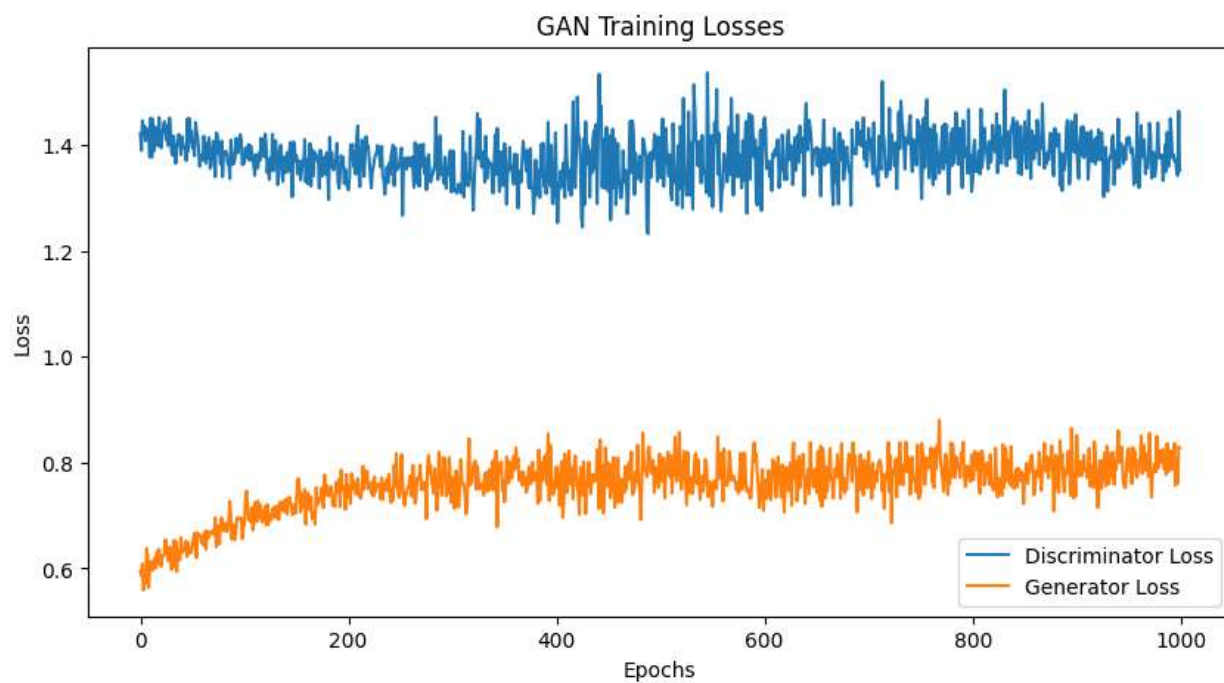
GAN DIAGRAM



GAN-Generator Unit Diagram



GAN-Discriminator Unit Diagram



RIGHT STEM - GROUP1 – Augmented by GAN Network data result:

coppre voxelpre zpre BSCpre

0	189.10	65.0	7.086	0.200
1	193.69	61.0	3.747	0.200
2	181.00	57.0	4.099	0.115
3	116.30	55.0	4.750	0.160
4	126.70	76.0	4.950	0.200
5	173.60	53.0	4.330	0.250
6	126.80	64.0	3.810	0.160
7	75.39	73.0	3.761	0.180
8	154.23	62.0	3.660	0.240
9	108.00	74.0	4.507	0.150
10	181.10	46.0	2.830	0.196
11	150.54	61.0	4.092	0.176
12	138.75	57.0	2.766	0.222
13	97.11	78.0	4.168	0.216
14	160.04	78.0	6.349	0.173
15	100.88	60.0	5.005	0.208
16	157.72	72.0	4.357	0.200
17	156.99	66.0	5.747	0.191
18	126.73	77.0	3.730	0.154
19	156.02	72.0	3.785	0.153
20	112.36	72.0	2.469	0.162
21	202.70	61.0	3.706	0.155
22	113.60	64.0	5.498	0.216
23	137.71	71.0	3.423	0.176
24	153.21	46.0	4.094	0.203
25	103.95	50.0	4.998	0.192
26	141.47	54.0	4.421	0.186
27	198.31	73.0	3.512	0.183
28	84.53	68.0	5.175	0.199
29	114.09	68.0	3.291	0.247
30	166.40	63.0	4.520	0.122
31	198.41	72.0	4.552	0.181
32	173.60	58.0	4.487	0.173
33	129.98	72.0	4.585	0.125
34	140.47	71.0	4.315	0.205
35	172.74	78.0	3.189	0.171
36	177.98	77.0	3.629	0.191
37	148.16	60.0	4.633	0.134
38	148.71	75.0	3.716	0.145
39	154.01	64.0	4.335	0.123

RIGHT STEM - GROUP2:

0	186.200	57	9.5020	0.1540
1	44.550	57	0.9130	0.8000
2	92.200	56	3.9590	0.1000
3	125.300	65	4.5590	0.1200
4	45.520	56	2.4310	0.0930
5	299.300	59	4.4850	0.1000
6	171.700	66	5.5680	0.2000
7	29.070	69	1.6470	0.1000
8	181.900	56	2.6150	0.0250
9	57.190	60	3.2320	0.0550
10	44.390	63	2.1940	0.0500
11	74.730	60	3.7860	0.0650
12	105.400	55	2.3130	0.0375
13	74.500	67	3.6910	0.0900
14	112.800	59	6.9360	0.1100
15	97.250	63	3.0290	0.0800
16	78.650	57	3.0650	0.1000
17	122.700	60	2.1080	0.0450
18	6.849	50	0.1419	0.1400
19	126.300	66	6.8990	0.2300
20	135.200	50	4.9120	0.2000
21	101.500	53	4.6330	0.0800
22	59.850	61	2.2950	0.0500
23	183.000	58	4.4480	0.1000
24	89.190	60	4.2140	0.0500
25	83.360	63	2.6420	0.0650
26	77.160	57	3.9520	0.1430
27	57.890	65	3.7810	0.1800
28	71.110	69	2.6810	0.2500
29	179.100	63	7.7710	0.2600
30	134.000	63	3.7520	0.0700
31	37.470	65	1.9020	0.0500
32	65.390	69	3.3750	0.1550
33	38.550	57	2.4290	0.0500
34	81.260	63	3.8240	0.1300
35	52.540	58	2.5650	0.0480
36	77.800	54	3.9240	0.1350
37	149.300	64	5.7850	0.2000
38	88.590	62	4.4770	0.0780

RIGHT STEM - GROUP1-Augmented vs GROUP2 T-Test Report:

T-test for cop:

t-statistic: 4.582079295271485

p-value: 1.745892758673924e-05

T-test for voxel:

t-statistic: 3.187417358471499

p-value: 0.00207505127019278

T-test for z:

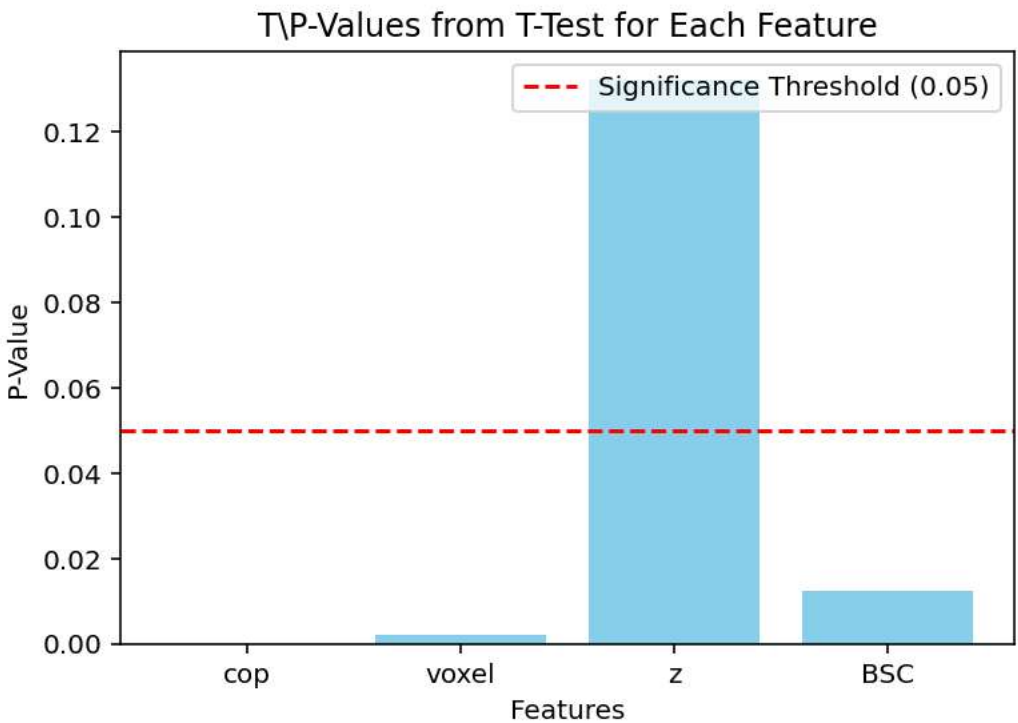
t-statistic: 1.5212140992126864

p-value: 0.13230225387940453

T-test for BSC:

t-statistic: 2.557337731018103

p-value: 0.012513091640255913



RIGHT STEM - GROUP1 (Original) vs GROUP2 T-Test Result:

T-test for cop:

t-statistic: 2.4075390820448495

p-value: 0.020039213984268618

T-test for voxel:

t-statistic: 1.798756202107894

p-value: 0.07848044508365437

T-test for z:

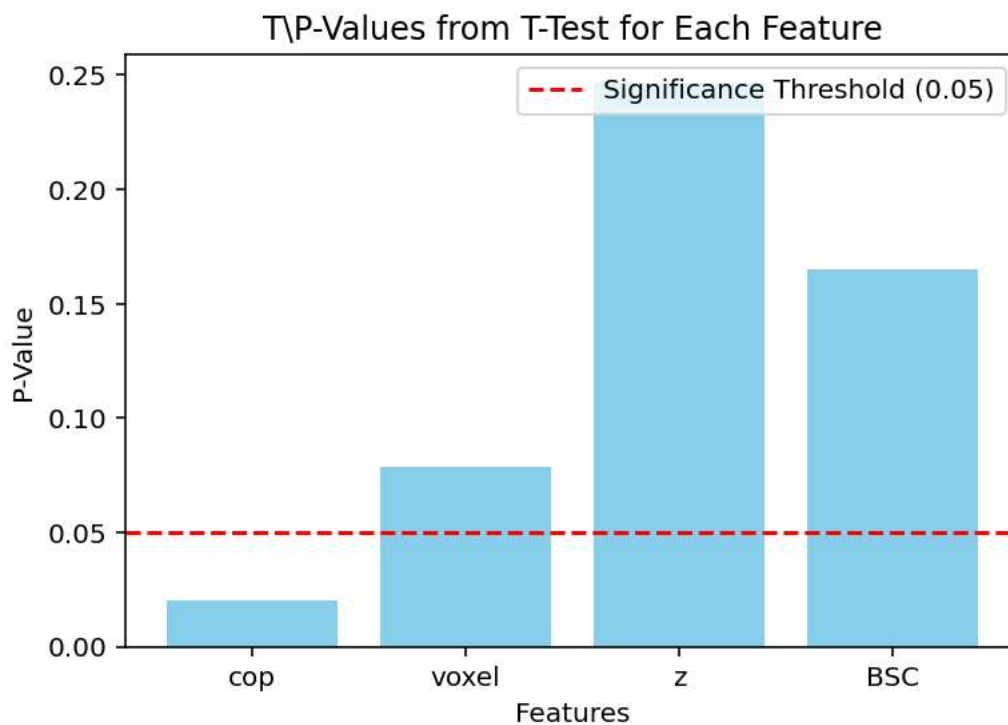
t-statistic: 1.1732315546383758

p-value: 0.24661459131521732

T-test for BSC:

t-statistic: 1.4112706347768293

p-value: 0.1647505622707071



RIGHT STEM - GROUP1-Augmented Statistics:

Mean coppre G1-A: 146.08

Mean voxelpre G1-A: 65.6

Mean zpre G1-A: 4.25

Mean BSCpre G1-A: 0.18

STDEV coppre: 32.31

STDEV voxelpre: 8.87

STDEV zpre: 0.91

STDEV BSCpre: 0.03

RIGHT STEM - GROUP1 Statistics:

Mean coppre: 144.48

Mean voxelpre: 64.0

Mean zpre: 4.47

Mean BSCpre: 0.19

STDEV coppre: 37.7

STDEV voxelpre: 7.68

STDEV zpre: 0.97

STDEV BSCpre: 0.04

RIGHT STEM - GROUP2 Statistics:

Mean cop G2: 98.43

Mean voxel G2: 60.38

Mean z G2: 3.75

Mean BSC G2: 0.13

STDEV cop G2: 56.08

STDEV voxel G2: 4.86

STDEV z G2: 1.82

STDEV BSC G2: 0.12

LEFT STEM - GROUP1 – Augmented by GAN Network data result:

	copp	voxel	zpre	BSCpre
0	119.000	60.000	5.438	0.120
1	159.110	60.000	4.600	0.190
2	181.000	51.000	5.023	0.125
3	212.800	58.000	4.208	0.100
4	133.200	70.000	4.355	0.200
5	156.240	48.000	5.710	0.150
6	222.900	58.000	4.900	0.160
7	147.900	65.000	4.780	0.150
8	200.300	60.000	4.420	0.100
9	208.000	72.000	6.510	0.200
10	116.029	52.875	4.507	0.090
11	142.854	56.683	6.070	0.133
12	115.388	56.332	5.947	0.143
13	157.858	84.754	5.598	0.141
14	188.978	48.209	6.110	0.155
15	172.311	55.817	5.903	0.174
16	172.629	69.254	5.714	0.167
17	152.307	51.494	6.334	0.164
18	152.217	69.109	4.614	0.152
19	216.684	69.446	4.757	0.169
20	142.363	52.023	4.932	0.130
21	134.150	59.185	4.956	0.111
22	175.693	63.070	5.201	0.176
23	194.514	51.652	4.429	0.193
24	150.879	58.139	5.614	0.128
25	136.992	64.006	5.249	0.124
26	159.289	57.999	4.918	0.183
27	204.219	62.571	5.042	0.097
28	199.806	66.114	5.176	0.105
29	122.133	70.493	3.581	0.182
30	214.894	49.191	4.792	0.148
31	186.978	59.783	4.228	0.139
32	206.946	60.868	4.489	0.138
33	196.887	66.322	5.308	0.132
34	186.113	66.366	4.293	0.158
35	178.298	53.134	4.366	0.106
36	142.368	51.148	5.644	0.161
37	65.987	49.870	4.690	0.126
38	191.583	57.693	4.396	0.144
39	154.613	61.407	5.079	0.136

LEFT STEM – GROUP2:

	copp	voxel	pre	zpre	BSCpre
0	168.20	53.0	7.7180	0.150	
1	44.55	56.0	0.6312	0.025	
2	148.20	64.0	2.5460	0.150	
3	130.40	53.0	5.1000	0.102	
4	68.58	50.0	3.0670	0.150	
5	359.50	53.0	4.3490	0.105	
6	171.70	62.0	5.5680	0.010	
7	96.86	59.0	4.5140	0.100	
8	201.90	55.0	3.8100	0.020	
9	71.07	57.0	3.5360	0.050	
10	66.27	65.0	3.2700	0.050	
11	78.24	55.0	3.6810	0.250	
12	105.40	55.0	3.7600	0.037	
13	75.90	64.0	3.0810	0.180	
14	250.50	55.0	6.4960	0.200	
15	108.20	58.0	5.0340	0.170	
16	50.10	61.0	3.1920	0.050	
17	122.70	53.0	2.1080	0.100	
18	43.48	60.0	1.8810	0.070	
19	226.30	60.0	6.7410	0.200	
20	207.10	48.0	4.7650	0.120	
21	169.90	53.0	7.7910	0.175	
22	59.85	59.0	3.3200	0.050	
23	221.10	58.0	3.5130	0.090	
24	89.19	55.0	2.5380	0.070	
25	141.30	56.0	3.6690	0.070	
26	105.70	50.0	5.0020	0.200	
27	63.03	59.0	3.4150	0.200	
28	125.00	64.0	3.5630	0.100	
29	168.60	58.0	5.6170	0.200	
30	90.83	61.0	3.3960	0.100	
31	38.08	58.0	1.7570	0.055	
32	54.89	65.0	2.8000	0.055	
33	63.89	51.0	4.0260	0.060	
34	111.20	61.0	4.9200	0.110	
35	52.96	51.0	2.2890	0.050	
36	104.50	50.0	5.4430	0.090	
37	197.80	59.0	1.8560	0.075	
38	111.70	60.0	4.1080	0.060	

LEFT STEM - GROUP1-Augmented vs GROUP2 T-Test Report:

T-test for coppre:

t-statistic: 3.62

p-value: 0.00052

T-test for voxelpre:

t-statistic: 1.990

p-value: 0.0501

T-test for zpre:

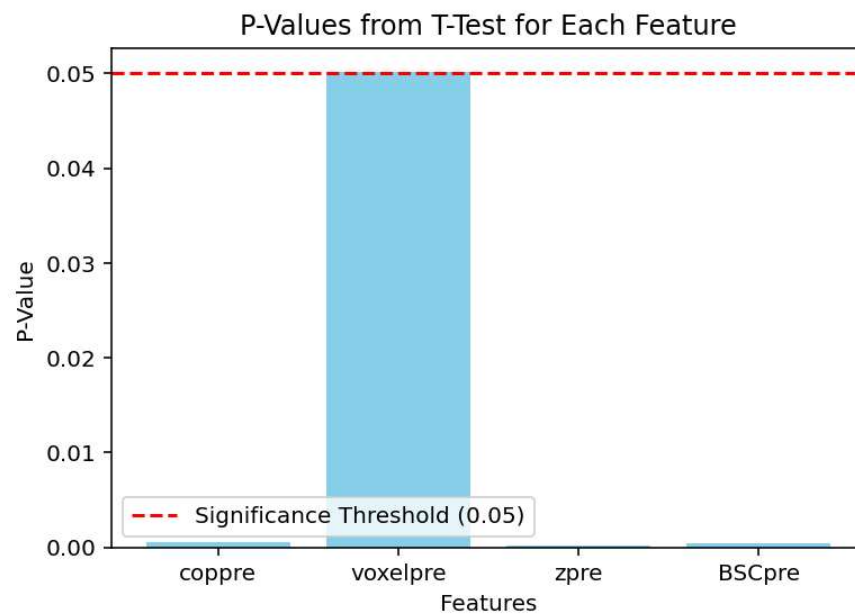
t-statistic: 4.02

p-value: 0.000133

T-test for BSCpre:

t-statistic: 3.69

p-value: 0.000413



LEFT STEM - GROUP1 (Original) vs GROUP2 T-Test Result:

T-test for coppre:

t-statistic: 2.267

p-value: 0.0280

T-test for voxelpre:

t-statistic: 1.703

p-value: 0.095

T-test for zpre:

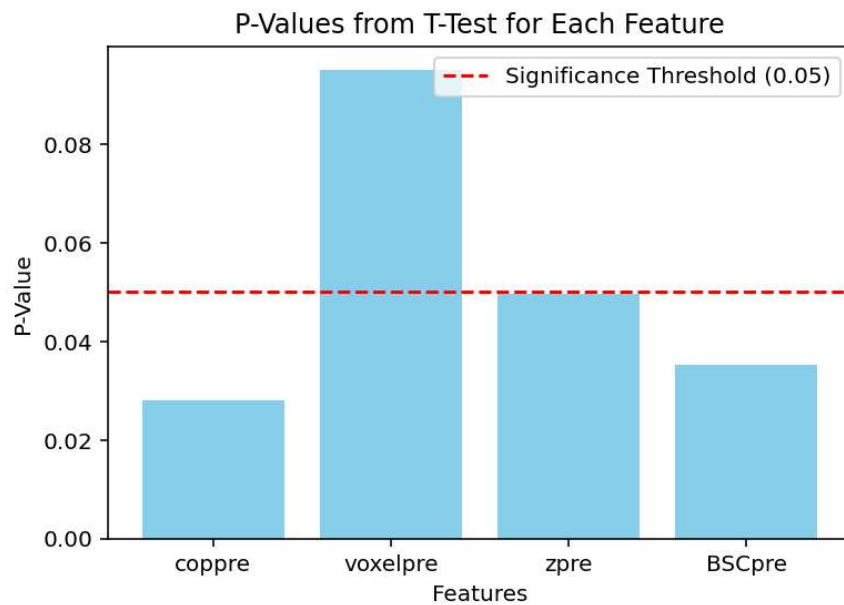
t-statistic: 2.015

p-value: 0.0496

T-test for BSCpre:

t-statistic: 2.166

p-value: 0.035



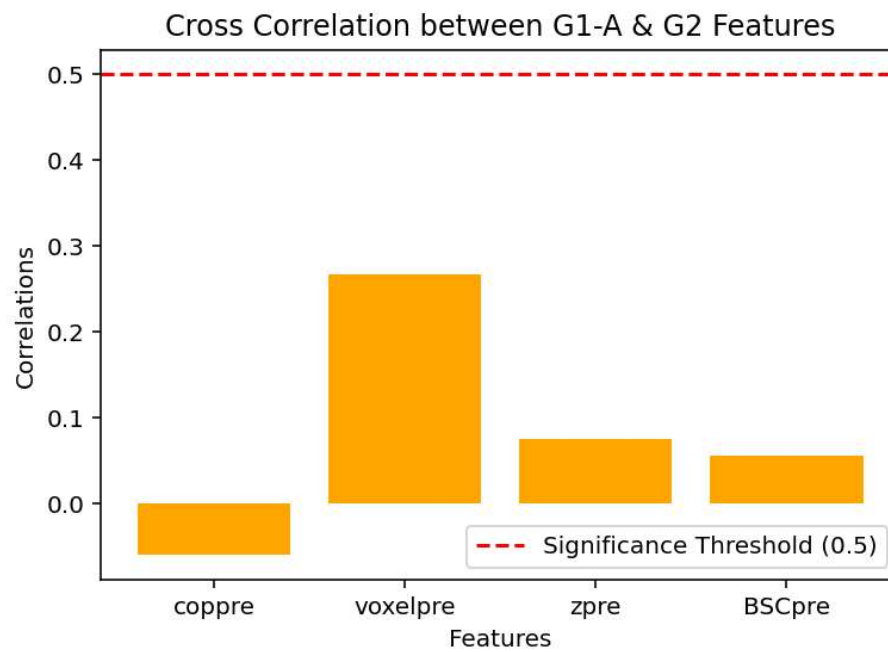
LEFT STEM - Correlation [GROUP1-Augmented vs GROUP2] :

Correlation coppre: -0.059

Correlation voxelpre: 0.2669

Correlation zpre: 0.074

Correlation BSCpre: 0.055



LEFT STEM - GROUP1-Augmented Statistics:

Mean coppre G1-A: 167.12

Mean voxelpre G1-A: 59.89

Mean zpre G1-A: 5.05

Mean BSCpre G1-A: 0.15

STDEV coppre G1-A: 34.67

STDEV voxelpre G1-A: 7.89

STDEV zpre G1-A: 0.67

STDEV BSCpre G1-A: 0.03

LEFT STEM - GROUP1 Statistics:

Mean coppre G1: 174.04

Mean voxelpre G1: 60.2

Mean zpre G1: 4.99

Mean BSCpre G1: 0.15

STDEV coppre G1: 34.21

STDEV voxelpre G1: 7.08

STDEV zpre G1: 0.68

STDEV BSCpre G1: 0.04

LEFT STEM - GROUP2 Statistics:

Mean coppre G2: 122.17

Mean voxelpre G2: 57.03

Mean zpre G2: 3.95

Mean BSCpre G2: 0.11

STDEV coppre G2: 68.71

STDEV voxelpre G2: 4.52

STDEV zpre G2: 1.58

STDEV BSCpre G2: 0.06

Normality test with Shapiro-Wilk Test:

RIGHT STEM GROUP1 - coppre:

Shapiro-Wilk test statistic: 0.9347502863043512

P-value: 0.4962041637397511

Data looks like a normal distribution

RIGHT STEM GROUP1 - voxelpre:

Shapiro-Wilk test statistic: 0.9317832509786342

P-value: 0.4656600163517724

Data looks like a normal distribution

RIGHT STEM GROUP1 - zpre:

Shapiro-Wilk test statistic: 0.7550868371782665

P-value: 0.004115879333578123

Data does not look like a normal distribution

RIGHT STEM GROUP1 - BSCpre:

Shapiro-Wilk test statistic: 0.9609977486590815

P-value: 0.797187667496022

Data looks like a normal distribution

RIGHT STEM GROUP1 Augmented - coppre:

Shapiro-Wilk test statistic: 0.9787502959937275

P-value: 0.6432029747873634

Data looks like a normal distribution

RIGHT STEM GROUP1 Augmented - voxelpre:

Shapiro-Wilk test statistic: 0.9468120674994568

P-value: 0.05893671039559701

Data looks like a normal distribution

RIGHT STEM GROUP1 Augmented - zpre:

Shapiro-Wilk test statistic: 0.9547641239869776

P-value: 0.11067094925897186

Data looks like a normal distribution

RIGHT STEM GROUP1 Augmented - BSCpre:

Shapiro-Wilk test statistic: 0.9790087708317297

P-value: 0.6527620151038658

Data looks like a normal distribution

RIGHT STEM GROUP2 - cop:

Shapiro-Wilk test statistic: 0.9080810749682929

P-value: 0.0037925736582603824

Data does not look like a normal distribution

RIGHT STEM GROUP2 - voxel:

Shapiro-Wilk test statistic: 0.9727908619783476

P-value: 0.4553839463451103

Data looks like a normal distribution

RIGHT STEM GROUP2 - z:

Shapiro-Wilk test statistic: 0.937613050807811

P-value: 0.03175035106647424

Data does not look like a normal distribution

RIGHT STEM GROUP2 - BSC:

Shapiro-Wilk test statistic: 0.579348565429754

P-value: 2.2356725074763545e-09

Data does not look like a normal distribution

LEFT STEM GROUP1 - coppre:

Shapiro-Wilk test statistic: 0.9415758512786505

P-value: 0.5706781090100408

Data looks like a normal distribution

LEFT STEM GROUP1 - voxelpre:

Shapiro-Wilk test statistic: 0.9444611265299422

P-value: 0.6036315202360633

Data looks like a normal distribution

LEFT STEM GROUP1 - zpre:

Shapiro-Wilk test statistic: 0.9110216629160103

P-value: 0.2880827649566656

Data looks like a normal distribution

LEFT STEM GROUP1 - BSCpre:

Shapiro-Wilk test statistic: 0.9079759242863261

P-value: 0.26737515719357663

Data looks like a normal distribution

LEFT STEM GROUP1 Augmented - coppre:

Shapiro-Wilk test statistic: 0.9615609841992756

P-value: 0.20099180597729982

Data looks like a normal distribution

LEFT STEM GROUP1 Augmented - voxelpre:

Shapiro-Wilk test statistic: 0.9487451710201183

P-value: 0.07455498596278698

Data looks like a normal distribution

LEFT STEM GROUP1 Augmented - zpre:

Shapiro-Wilk test statistic: 0.9710103138586211

P-value: 0.4031273395836985

Data looks like a normal distribution

LEFT STEM GROUP1 Augmented - BSCpre:

Shapiro-Wilk test statistic: 0.9776450325797402

P-value: 0.6182274923221711

Data looks like a normal distribution

LEFT STEM GROUP2 - coppre:

Shapiro-Wilk test statistic: 0.8956698870973706

P-value: 0.001661701624117287

Data does not look like a normal distribution

LEFT STEM GROUP2 - voxelpre:

Shapiro-Wilk test statistic: 0.9680980893792391

P-value: 0.32765361461131115

Data looks like a normal distribution

LEFT STEM GROUP2 - zpre:

Shapiro-Wilk test statistic: 0.9670293946306404

P-value: 0.30305369562342976

Data looks like a normal distribution

LEFT STEM GROUP2 - BSCpre:

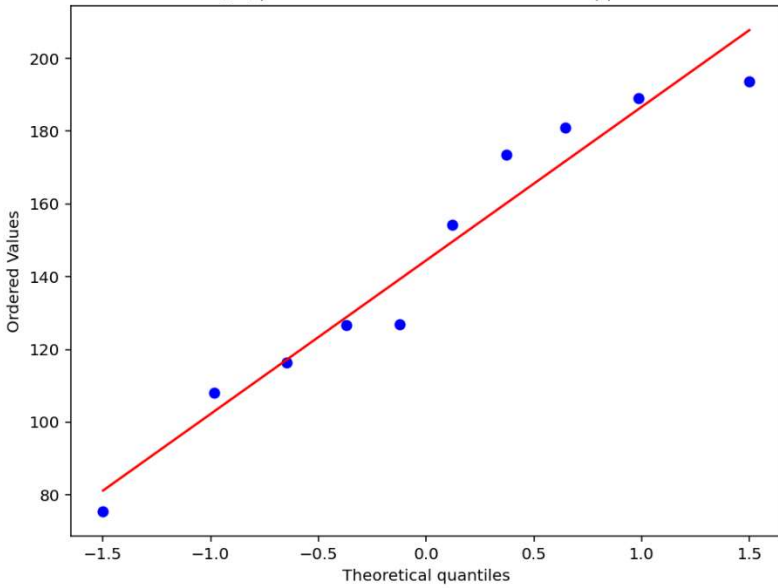
Shapiro-Wilk test statistic: 0.9293010855318422

P-value: 0.017067319779668567

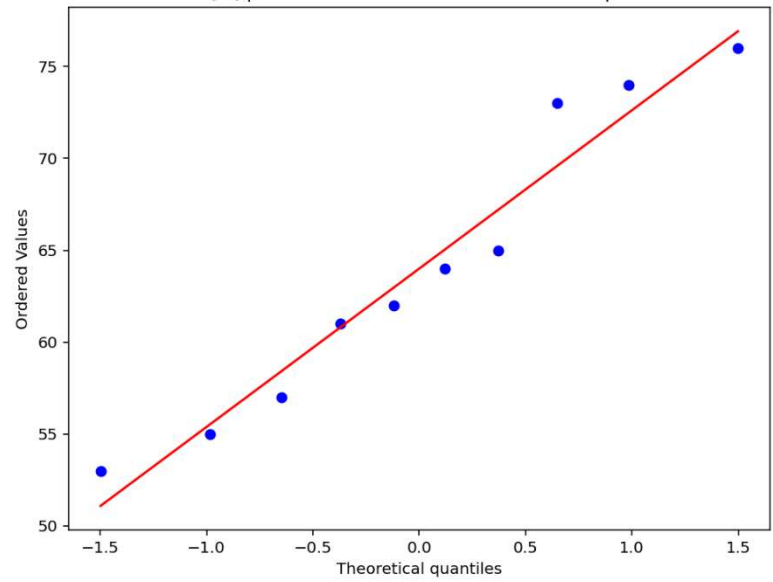
Data does not look like a normal distribution

Q-Q Visual Test for normality test – STEM RIGHT:

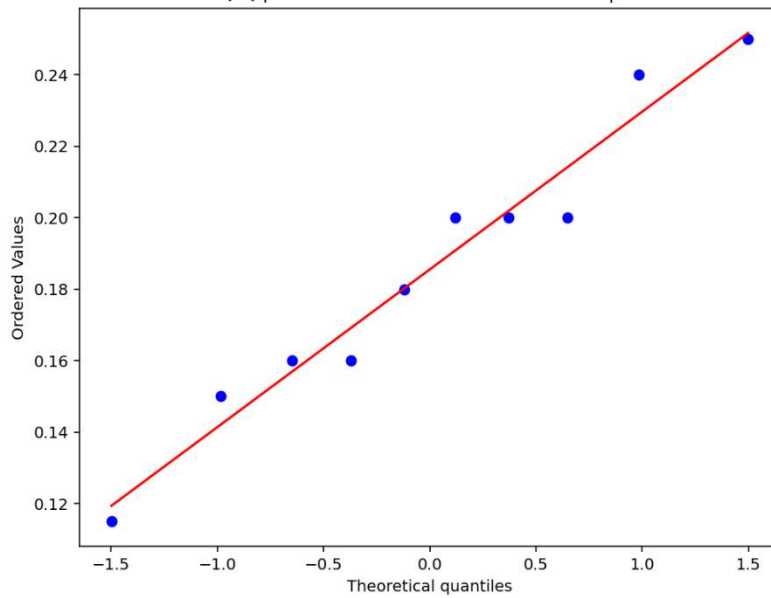
Q-Q plot for RIGHT STEM - GROUP1 - coppre



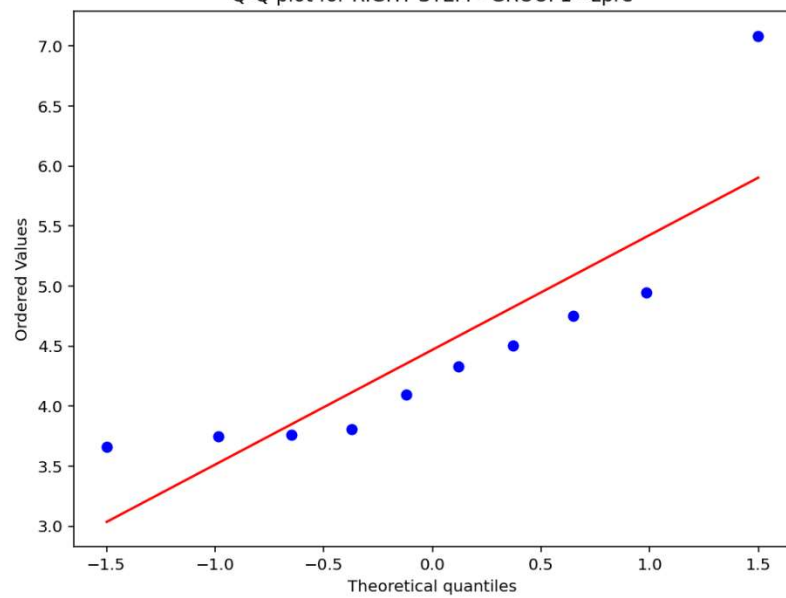
Q-Q plot for RIGHT STEM - GROUP1 - voxelpre



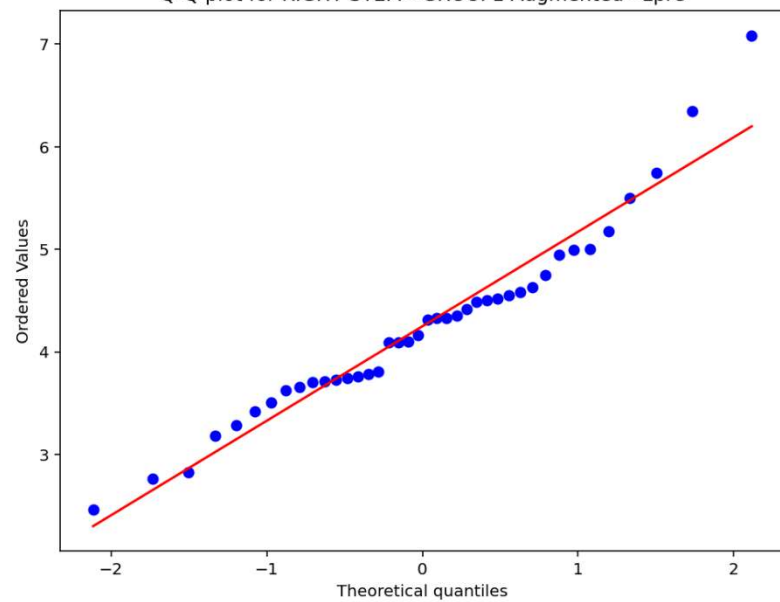
Q-Q plot for RIGHT STEM - GROUP1 - BSCpre



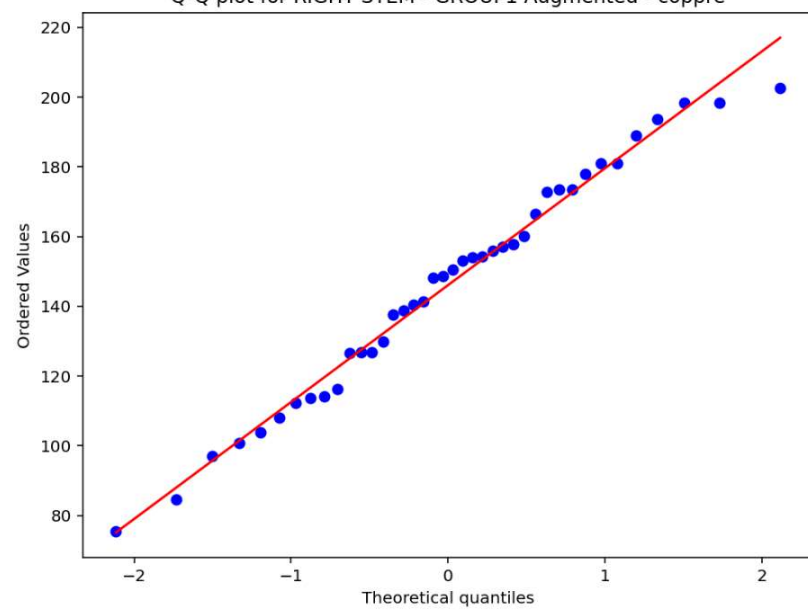
Q-Q plot for RIGHT STEM - GROUP1 - zpre



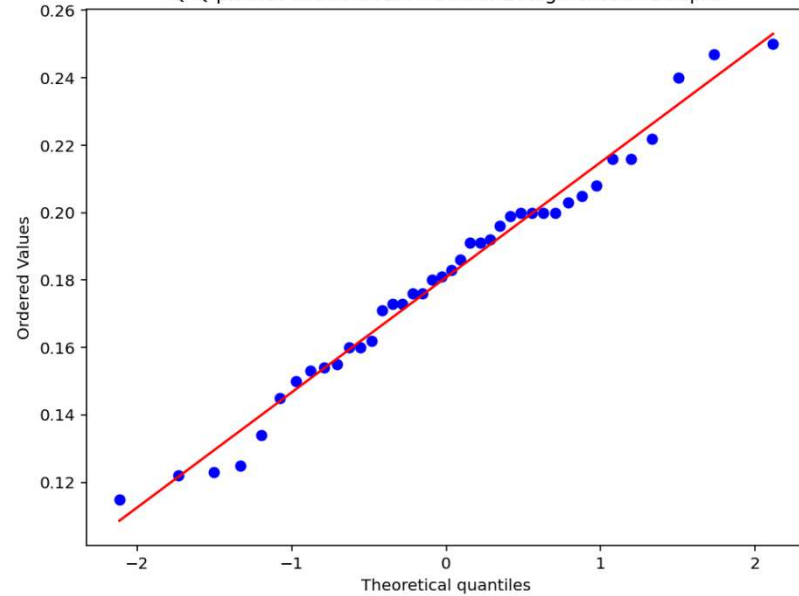
Q-Q plot for RIGHT STEM - GROUP1 Augmented - zpre



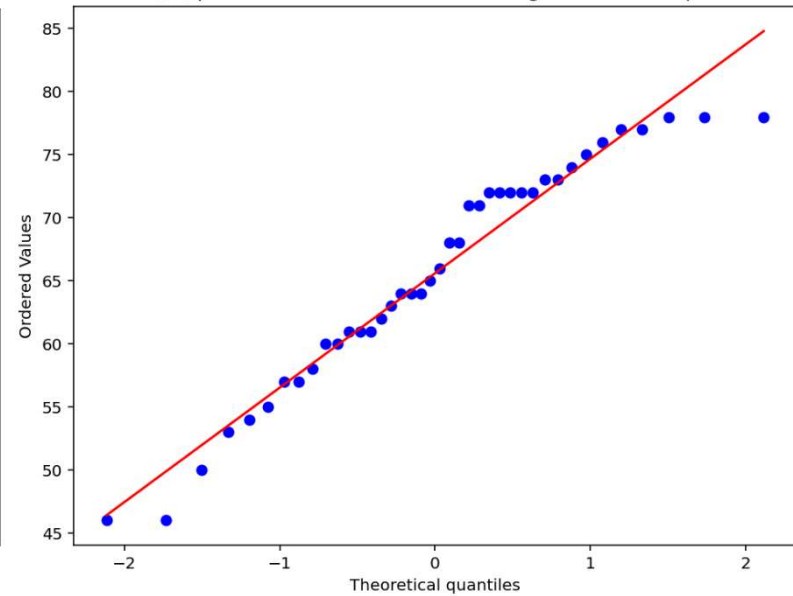
Q-Q plot for RIGHT STEM - GROUP1 Augmented - coppre



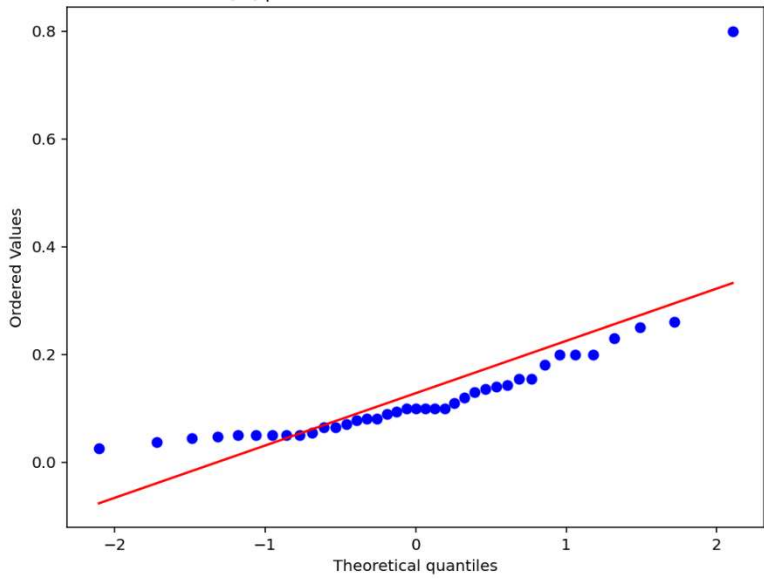
Q-Q plot for RIGHT STEM - GROUP1 Augmented - BSCpre



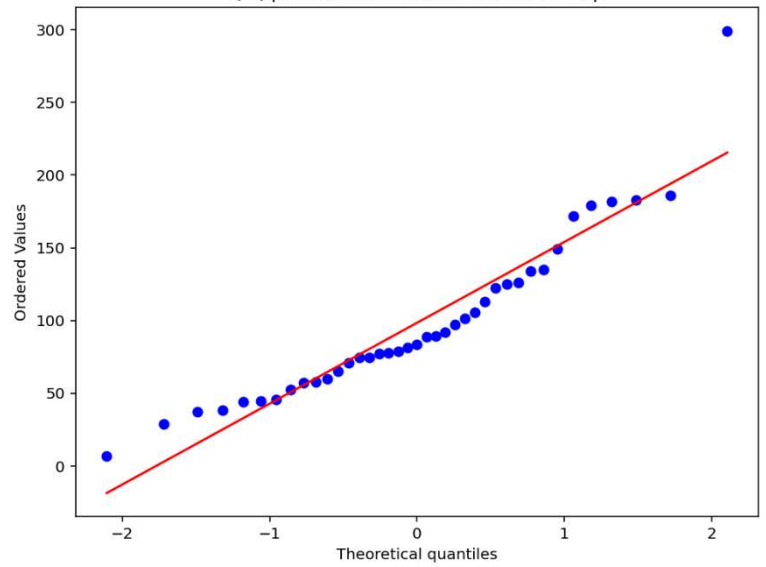
Q-Q plot for RIGHT STEM - GROUP1 Augmented - voxelpre



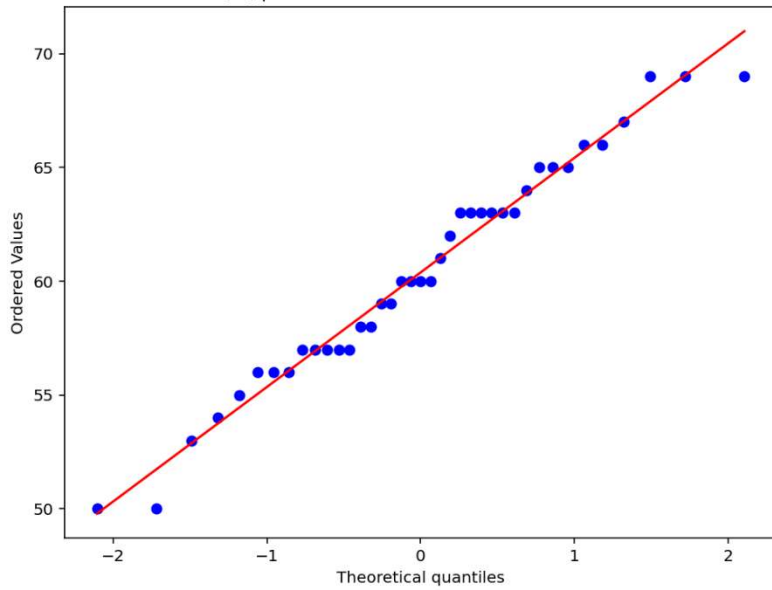
Q-Q plot for RIGHT STEM - GROUP2 - BSC



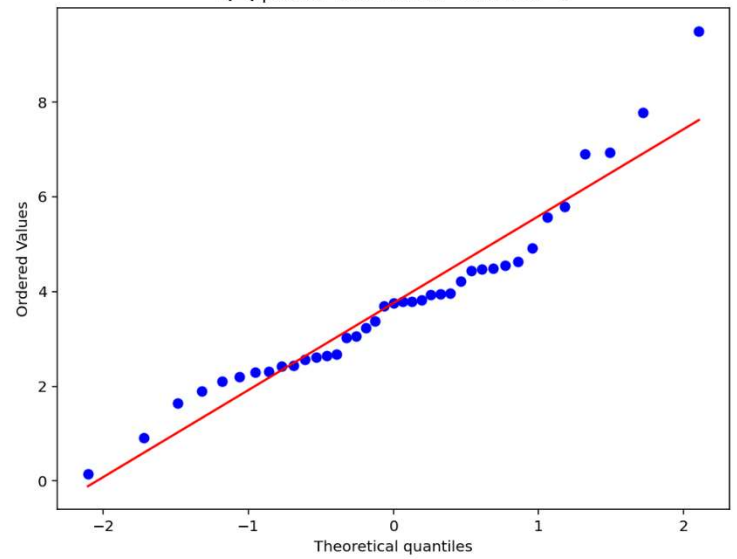
Q-Q plot for RIGHT STEM - GROUP2 - cop



Q-Q plot for RIGHT STEM - GROUP2 - voxel



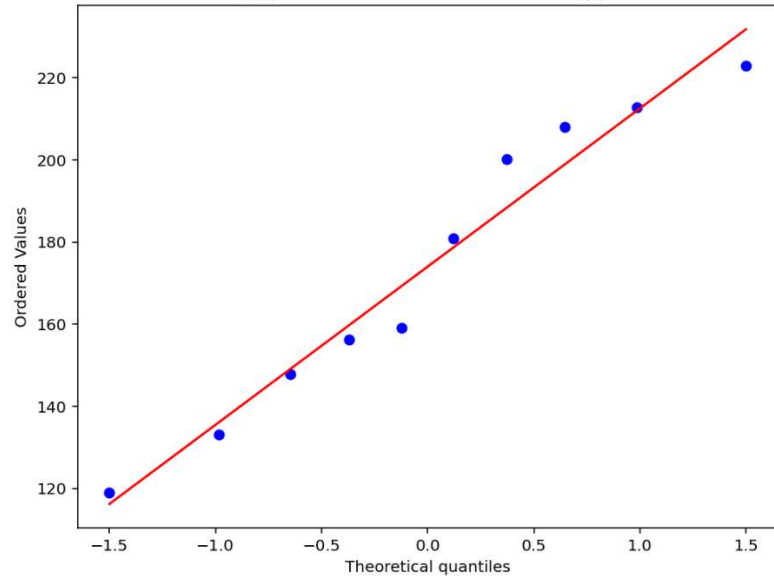
Q-Q plot for RIGHT STEM - GROUP2 - z



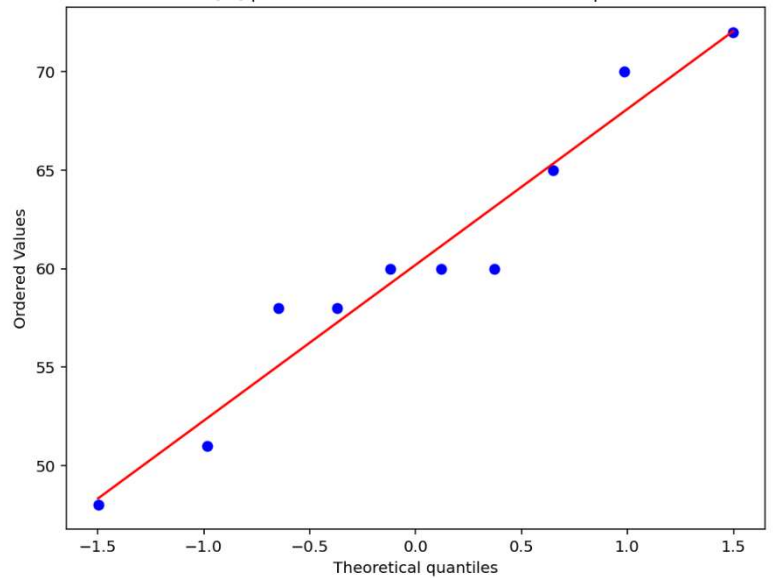
PLOTS ON THIS PAGE CHANED

Q-Q Visual Test for normality test – STEM LEFT:

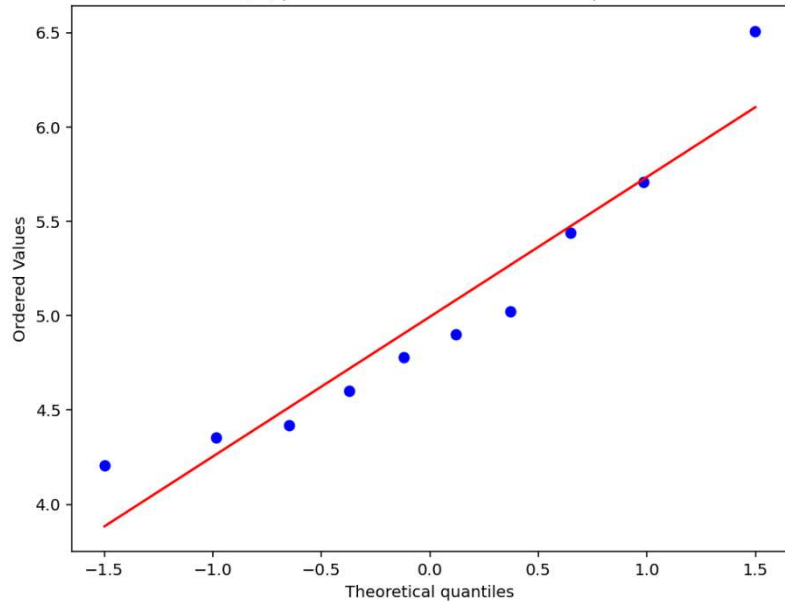
Q-Q plot for LEFT STEM - GROUP1 - coppre



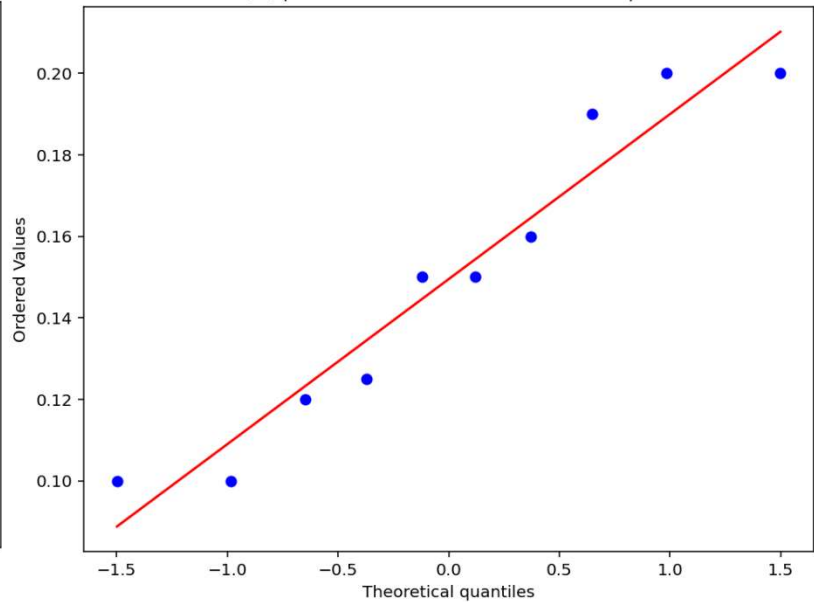
Q-Q plot for LEFT STEM - GROUP1 - voxelpre



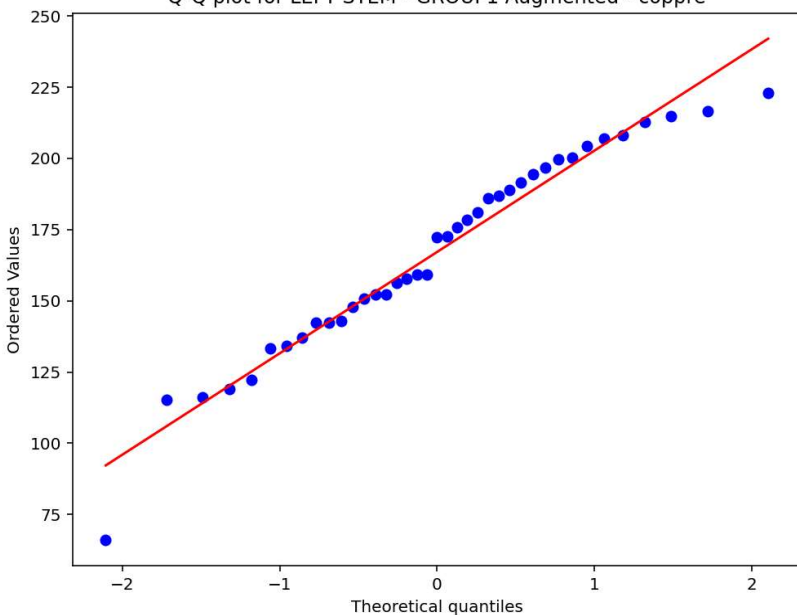
Q-Q plot for LEFT STEM - GROUP1 - zpre



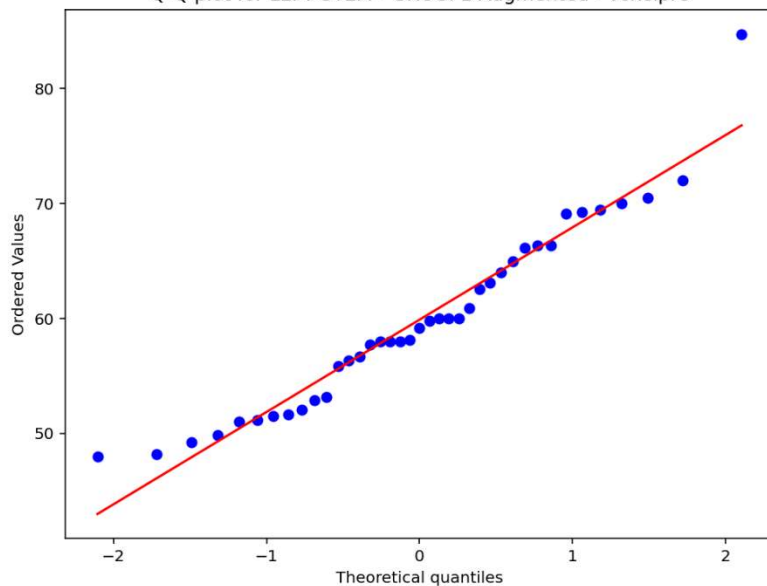
Q-Q plot for LEFT STEM - GROUP1 - BSCpre



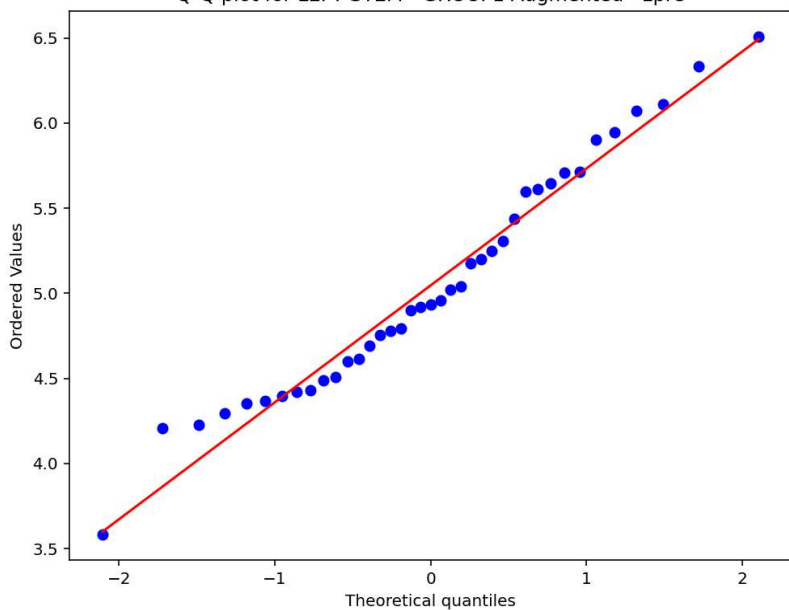
Q-Q plot for LEFT STEM - GROUP1 Augmented - coppre



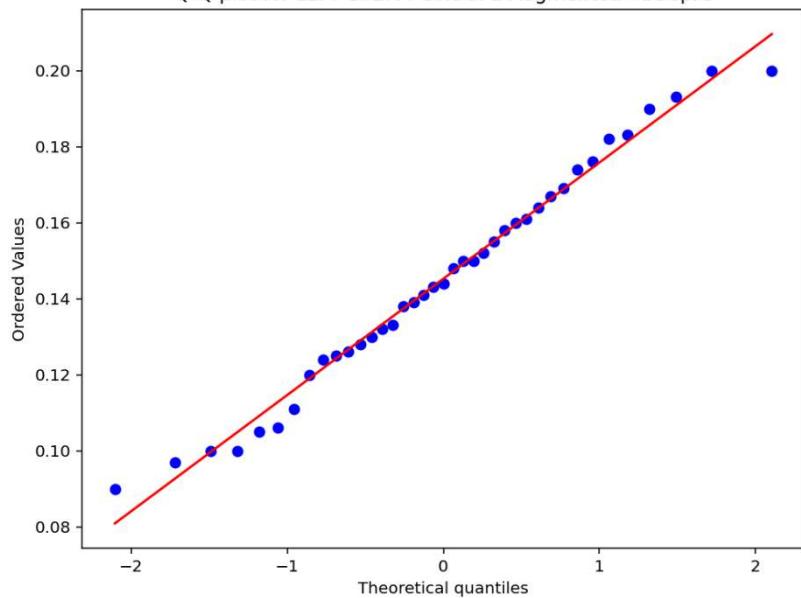
Q-Q plot for LEFT STEM - GROUP1 Augmented - voxelpre



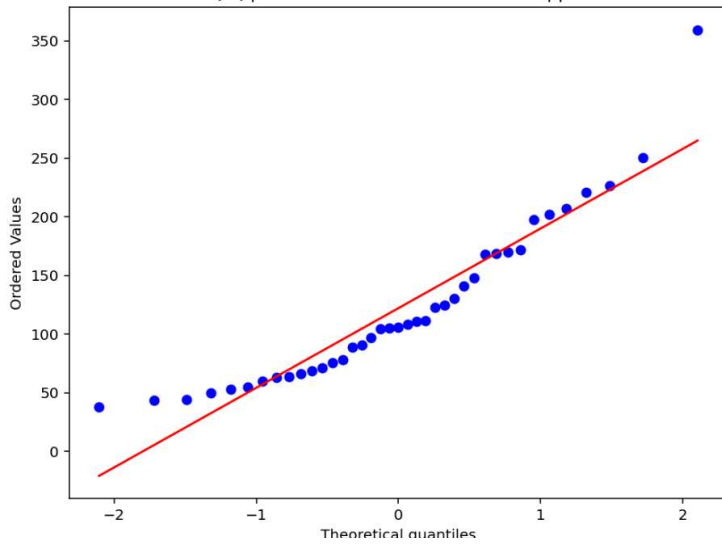
Q-Q plot for LEFT STEM - GROUP1 Augmented - zpre



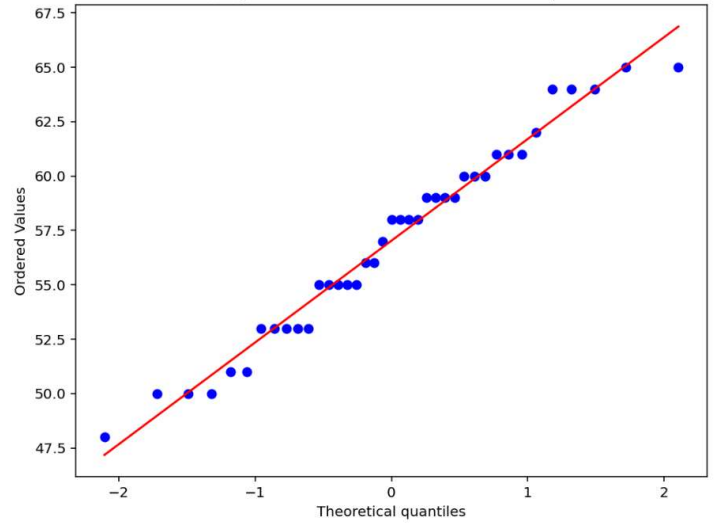
Q-Q plot for LEFT STEM - GROUP1 Augmented - BSCpre



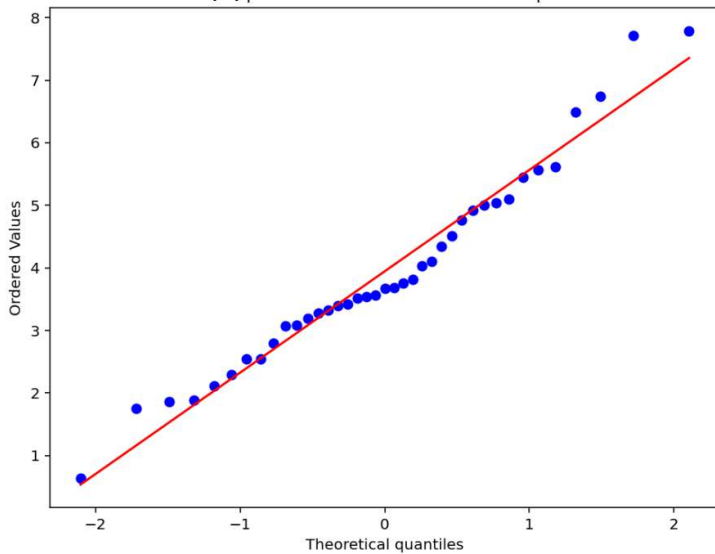
Q-Q plot for LEFT STEM - GROUP2 - coppre



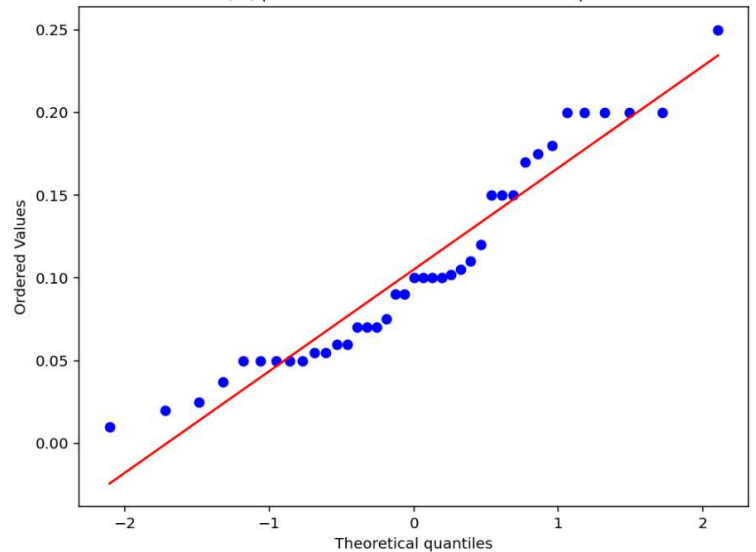
Q-Q plot for LEFT STEM - GROUP2 - voxelpre



Q-Q plot for LEFT STEM - GROUP2 - zpcre



Q-Q plot for LEFT STEM - GROUP2 - BSCpre



G1 vs G2 RIGHT ARI Assessment:

Adjusted Rand Index (ARI): 0.11023285622730186

G1-A vs G2 RIGHT ARI Assessment:

Adjusted Rand Index (ARI): 0.31586558212143623

G1 vs G2 LEFT ARI Assessment:

Adjusted Rand Index (ARI): 0.10715394968423232

G1-A vs G2 LEFT ARI Assessment:

Adjusted Rand Index (ARI): 0.3711814167322001

